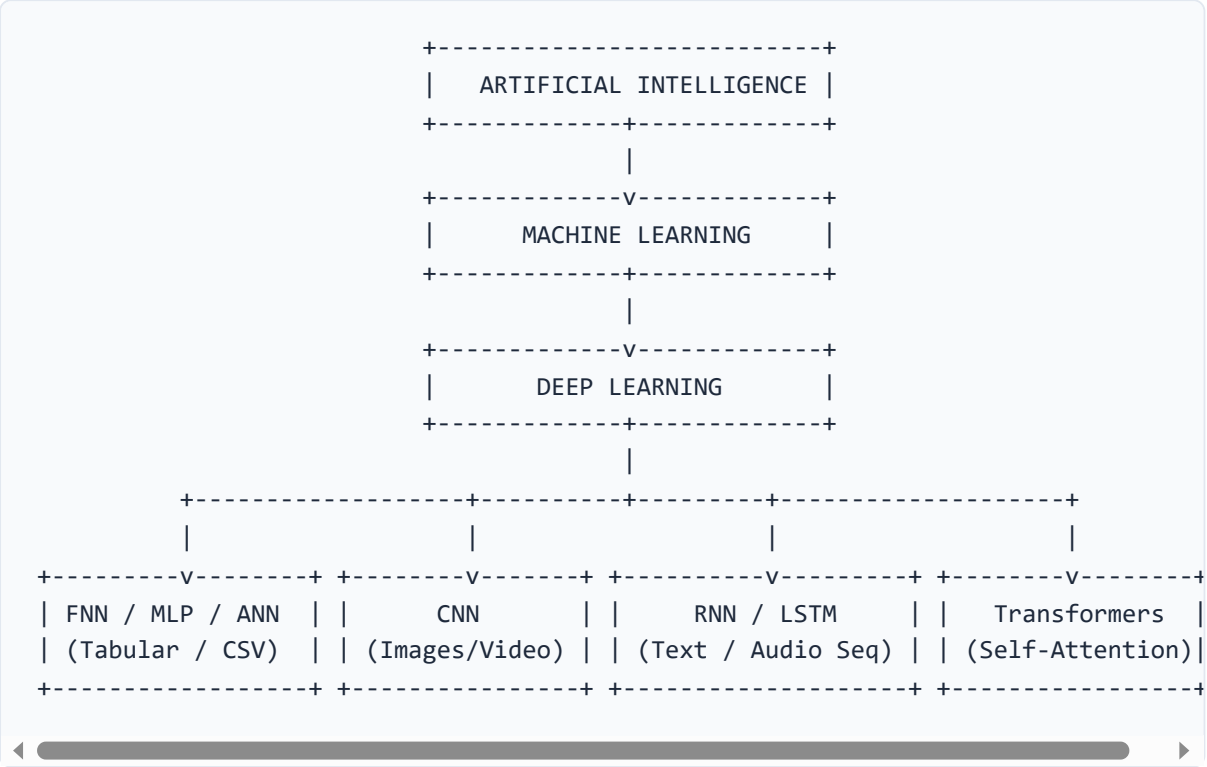
Submodule 39.01: Deep Learning Architecture Taxonomy & The Feedforward Paradigm

1. Introduction: The Deep Learning Family of Architectures

Deep Learning is a specialized subfield of Machine Learning based on multi-layered artificial neural networks. While all neural networks utilize interconnected nodes (artificial neurons) that learn representations via gradient descent and backpropagation, **different data modalities require specialized architectural inductive biases**.



---

2. Feedforward Neural Networks (FNN / MLP / ANN)

A **Feedforward Neural Network (FNN)**, also referred to as a **Multilayer Perceptron (MLP)** or standard **Artificial Neural Network (ANN)**, is the simplest and foundational deep learning architecture:

Key Architectural Characteristics:

1. **Unidirectional Information Flow**: Information moves strictly forward through the network—from the input layer, through one or more hidden layers, to the output layer:

$$\mathbf{x} \to \mathbf{h}_1 \to \mathbf{h}_2 \to \cdots \to \mathbf{h}_L \to \hat{\mathbf{y}}$$

2. **No Recurrent Loops**: Neurons in layer l receive connections only from layer $l-1$ and project only to layer $l+1$. There are no cycles, internal state feedback loops, or temporal memory.
3. **Clarifying Backpropagation in FNNs**:
    - A widespread student misconception is that backpropagation contradicts the "feedforward" definition.
    - **Forward Propagation** passes input data forward to compute predictions and loss.
    - **Backpropagation** merely passes mathematical error gradients ($\frac{\partial L}{\partial \mathbf{W}}$) backward using the chain rule to update learnable parameters. The underlying network data flow remains strictly feedforward.

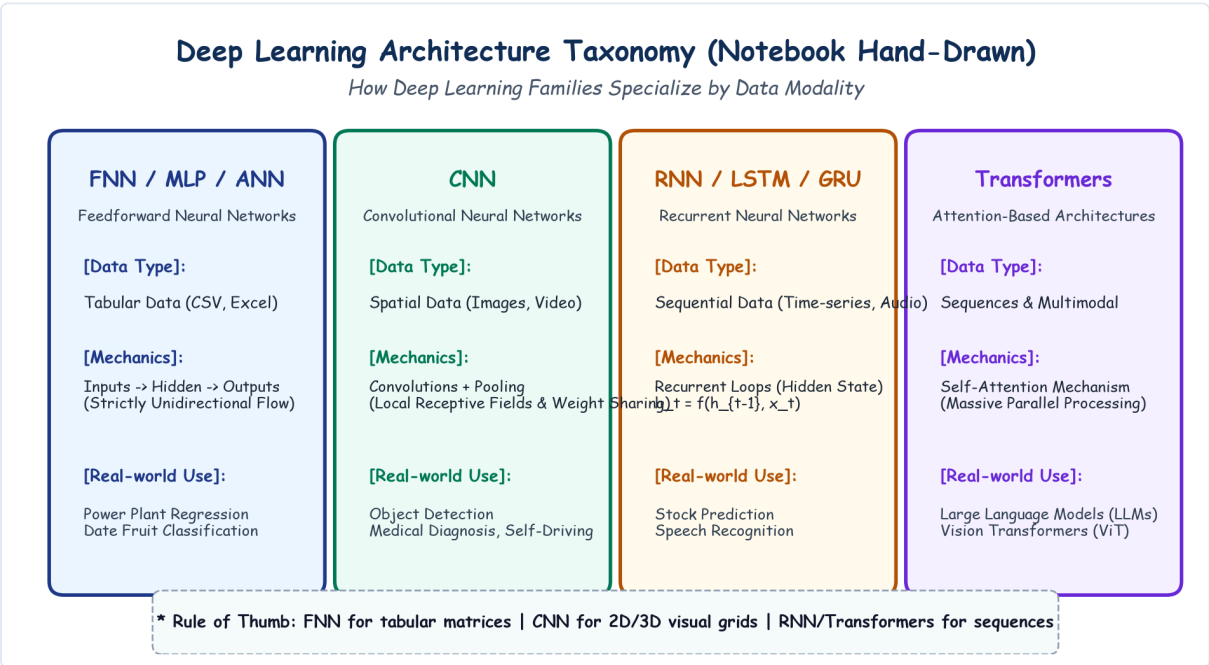
Primary Domain:

FNNs are optimized for **structured tabular data** (e.g. database rows, CSV files, Excel spreadsheets) where input features are scalar quantities without spatial or temporal adjacency.

---

3. Hand-Drawn Educational Diagram: Architecture Taxonomy

The following student-style notebook diagram illustrates how deep learning architectures specialize by data modality, showing their distinct wiring patterns.



---

4. Architectural Comparison Table

Architecture Family	Primary Data Modality	Core Computational Mechanism	Primary Industry Applications
FNN / MLP	Tabular / Structured Data	Dense Matrix Multiplication ($\mathbf{Wx} + \mathbf{b}$)	Credit scoring, churn prediction, tabular regression
CNN	Spatial 2D/3D Grids (Images, Video)	Sliding Convolutions & Spatial Pooling	Object detection, facial recognition, autonomous driving
RNN / LSTM / GRU	Sequential / Temporal Data	Recurrent State Transition ($\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$)	Time-series forecasting, speech recognition, ECG monitoring
Transformers	Sequences, Multimodal, Vision	Scaled Dot-Product Self-Attention	Large Language Models (LLMs), Vision Transformers (ViT)

---

5. Limitations of Feedforward Networks

While highly effective on tabular data where each column represents an independent feature (e.g., age, salary, ambient temperature), FNNs become catastrophically inefficient when applied to high-dimensional grid structures like digital images. In the following submodules, we explore why this occurs and how **Convolutional Neural Networks (CNNs)** solve it.